Urdu Sarcasm Detection

Mohammad Mohib Qureshi
RollNo:25K7617
NLP
Programming Assignment 1

83.4%	0.830	59.5%	14,568
accuracy (ensemble, test)	Macro-F1 (primary metric)	majority-class baseline	clean posts after cleaning

Summary. The best system is a weighted ensemble. It reaches **Macro-F1 0.830** and **accuracy 83.4%** on a held-out test set of 2,186 posts (a majority-class baseline gets 59.5%). All model families land between 0.807 and 0.830 Macro-F1. So label noise and missing conversational context, and not model size, are what limit the score. Macro-F1 is the primary metric because the classes are imbalanced (40.5% sarcastic).

1. Data preparation and exploratory analysis

Cleaning step	Rows removed	Rows left	Why
Raw file	—	20,060	starting point
Missing text or label	105	19,955	cannot train on them
Exact duplicates	4,095	15,860	avoid train/test leak
Same text, both labels (646 texts)	1,292	**14,568**	proof of label noise

After cleaning, 14,568 posts remain (8,661 non-sarcastic, 5,907 sarcastic). They were split 70/15/15 with stratification into 10,197 / 2,185 / 2,186 posts. Every vectorizer, tokenizer, vocabulary, phrase list and scaler is fit on the training split only. Thresholds and ensemble weights are tuned on validation only, and the test set is used once at the end.

Class profile	Non-sarcastic (0)	Sarcastic (1)	Sarcastic / non
Posts (share)	8,661 (59.5%)	5,907 (40.5%)	—
Words per post: mean (median)	15.7 (12)	23.9 (21)	1.5×
Characters per post (mean)	71.6	112.8	1.6×
Posts with a hashtag	0.3%	11.9%	39×
Mentions per post	0.014	0.167	12×
Posts with a URL	0.3%	5.8%	19×
Posts with Latin letters (code-mixing)	7.5%	27.6%	3.7×
Posts with any emoji	88.1%	66.8%	0.76×

Emoji cues cut both ways. Hearts, roses and flowers almost never appear in sarcastic posts (log-odds −2.2 to −3.5), while winking, tongue and grinning faces lean sarcastic (+0.6 to +0.8). The tears-of-joy face is the most frequent emoji in both classes (26.9% vs 23.1%), so it carries little signal alone.

Label leakage. 96.4% of posts with a hashtag are sarcastic, but the top tags are the label itself (#sarcasm 376, #150 طنز, #sarcastic 56, #21 مزاح, #satire 16). These tags are stripped from every model input. Other data-quality issues are spelling variation, Punjabi and English mixed into Urdu, emoji-only or very short posts that need a parent post, and several doubtful gold labels.

Representative examples

Class	Post (Urdu, as in the data)
Sarcastic	وزیر دفاع بیچارہ خود بیساکھیوں کے سہارے ہے 😂
Sarcastic	سندھ پولیس اگر کام نہیں کر سکتی تو انڈیا چلی جائے 🙄
Non-sarcastic	وہ تیر مجھے کیونکر نہ لگا جس تیر سے گھائل تھا اصغر ع 💔
Non-sarcastic	صرف مانع تھی حیا بند قبا کھلنے تلک پھر تو وہ جان حیا ایسا کھلا ایسا کھلا

2. Urdu-aware Preprocessing

Letter unification. Arabic and Urdu code points for the same letter are merged (ا←أ/إ/آ, ه←ہ, ک←ك, ی←ي). Diacritics and tatweel are removed because people type them inconsistently. The do-chashmi he (ھ) stays separate from ہ, because merging them changes real words (تھا would become تہا).

Cues become tokens and are not deleted. URLs, mentions and numbers become xxurl, xxuser and xxnum. Emojis are split into separate tokens. Elongation, “!” / “!!”, “؟؟” / “؟”, “...” and quotation marks become marker tokens. Hashtags become xxhash plus their words (minus the label-word tags). The assignment asks us not to remove these blindly, so each one is tested in the ablation.

Two views of the text. norm_full (all of the above) feeds TF-IDF and the BiLSTM. A minimal view (only masking, leak-tag removal and letter unification) feeds the char-CNN and XLM-R, because their character and subword vocabularies already handle raw emojis and punctuation.

Stopwords (131 words, Kaggle urdu-stopwords) are removed only as an ablation. Negations and intensifiers (نہیں، نہ، بہت، واہ، کیا، بس، ہی) are always kept, since ironic praise depends on them. There is no stemming, because English stemmers do not fit Urdu morphology.

3. Models, features and why each one was used

Model	Why we chose it	Setup
TF-IDF + LogReg / LinearSVM / ComplementNB	Fast, strong baseline for small data, and easy to interpret. **BPE was tried because Urdu has a very large vocabulary** with many spelling variants and word forms. A whole-word vocabulary misses many of them, while BPE cuts rare words into reusable pieces so the model can still catch them. Character n-grams help for the same reason.	BPE-4k (1–4-grams) + char 2–5-grams, 118,978 features. Class-weighted.
Phrases + handcrafted features (bonus)	Sarcasm often sits in fixed phrases and pragmatic cues, which plain n-grams split up. PMI collocations turn phrases into single tokens, and the handcrafted features add cues that words alone miss.	1,989 PMI phrases (PMI ≥ 3, count ≥ 5), e.g. السلام_علیکم, pak_army. 18 scaled features (length, emoji / laugh / mock counts, "!", "؟", ellipsis, quotes, elongation, Latin ratio, mentions, URL, negation, positive words) + an incongruity flag (positive word with a laughing or mocking emoji).
BiLSTM-attention + char-CNN (PyTorch)	An LSTM reads the words in order, and sarcasm depends on contrast between parts of a sentence. Attention points to the incongruous span. The char-CNN handles misspellings, and FastText embeddings share information between spelling variants.	2-layer BiLSTM (hidden 128) over FastText-CBOW vectors trained on the train split; attention + max-pooling; char-CNN kernels 3/4/5; handcrafted features concatenated. Last 80 tokens kept, since sarcasm markers cluster at the end. AdamW + OneCycle, class-weighted loss, early stopping.
XLM-RoBERTa-large (fine-tuned)	A multilingual transformer pretrained on 100 languages including Urdu, so it suits Urdu mixed with English and Roman words. Its SentencePiece vocabulary covers emojis and Latin subwords, and pretraining gives knowledge that 10k posts cannot.	lr 1e-5, 10% warm-up, effective batch 32, fp16 on a T4, max length 128, best of 4 epochs on validation.
Weighted ensemble	The models make different mistakes (the BiLSTM has high sarcastic recall, the linear model has high precision), so averaging them helps a little.	Probability average. Weights and threshold tuned on validation: FE-LogReg 0.62, BiLSTM 0.31, XLM-R 0.06; threshold 0.51.

4. Ablation study (validation Macro-F1, TF-IDF + LogReg)

Text variant	Word 1–3	Char 2–5	Word+char	BPE alone	BPE+char	Best
norm_full (all cues kept)	0.788	0.797	0.795	0.788	0.800	**0.800**
no emojis	0.776	0.792	0.792	0.777	0.795	0.795
no punctuation markers	0.783	0.793	0.793	0.790	0.798	0.798
no hashtags	0.786	0.794	0.789	0.786	0.794	0.794
stopwords removed	0.782	0.793	0.796	0.791	0.795	0.796
aggressive (all removed)	0.757	0.778	0.778	0.762	0.779	0.779

Every removed cue hurts. Aggressive cleaning costs 2.1 points, dropping emojis costs 1.2 points on word features, and stopword removal does not help because function words such as کیا and بس carry rhetorical tone. Character features beat word features in every variant, and BPE helps only when it is combined with characters. Phrases and handcrafted features add +0.7 points on validation (0.807 vs 0.800).

5. Results (test set, n = 2,186)

Model	Acc	P(0)	R(0)	F1(0)	P(1)	R(1)	F1(1)	Macro-F1
Weighted ensemble	0.834	0.875	0.842	0.858	0.781	0.823	0.801	**0.830**
TF-IDF + phrases + handcrafted (LR)	0.831	0.869	0.842	0.856	0.779	0.814	0.796	**0.826**
XLM-RoBERTa-large (fine-tuned)	0.830	0.869	0.842	0.855	0.778	0.814	0.795	**0.825**
BiLSTM-attn + char-CNN + features	0.829	0.886	0.817	0.850	0.759	0.845	0.800	**0.825**
TF-IDF LogReg (BPE4k + char)	0.827	0.861	0.846	0.853	0.780	0.799	0.789	**0.821**
TF-IDF LinearSVM (BPE4k + char)	0.826	0.864	0.840	0.852	0.774	0.806	0.790	**0.821**
TF-IDF ComplementNB (BPE4k + char)	0.817	0.825	0.879	0.851	0.803	0.726	0.762	**0.807**

Confusion matrices (rows = true label, columns = predicted label)

Weighted ensemble

true \ pred	non-sarc	sarc
non-sarc	1095	205
sarc	157	729

XLM-RoBERTa-large

true \ pred	non-sarc	sarc
non-sarc	1094	206
sarc	165	721

BiLSTM-attn + char-CNN

true \ pred	non-sarc	sarc
non-sarc	1062	238
sarc	137	749

The top four models are within 0.5 points of each other. McNemar's test between the ensemble and the FE-LogReg runner-up gives p = 0.53, so the ensemble gain is **not statistically significant**. XLM-R-large matches the feature-engineered linear model but does not beat it. About 10k training posts is little data for a 560M-parameter model, Urdu is a small share of its pretraining data, and the decisive cues (emoji type, mentions, URLs, length) are surface features that a linear model already captures. The BiLSTM has the best sarcastic-class recall (0.845) but pays for it with more false positives (238).

6. Error analysis (weighted ensemble, 10 misclassified posts)

The ensemble misclassifies 16.6% of test posts. The error rate is highest for long posts (22.8%), rhetorical questions (21.3%), posts with laughing or mocking emojis (21.0%) and negation (18.0%), and lowest for very short posts (6.4%). 178 test posts are missed by every model. These are the most confident errors (FN = missed sarcasm, FP = false alarm):

#	Type	Post (shortened)	Phenomenon	Commentary
1	FN	صلی اللہ علیہ وآلہ وسلّم🌹🌹🌹🌹🌹	Ambiguous / wrong label	A plain religious blessing with roses and no irony. This looks like a labelling error.
2	FN	خدا کرے.. تو میری یاد میں خاک چھانے … 🥀🌺🌹	Idiom / poetry + sincere emoji style	A couplet with flowers (score 0.015). The bitter wish is the irony, and seeing it needs knowledge of the verse.
3	FN	آپ بھی 🍃🍃🍃🍃	Context dependence	A very short reply. Any sarcasm depends on a parent post that is missing, so the label is ambiguous.
4	FN	تو میرا خیر خاہ 😍😂😂😂	Spelling variation	The non-standard spelling خاہ is an unseen word, and the praise-plus-laughter pattern is too short to learn from.
5	FN	وادہ وکہ 😂	Spelling variation	A phonetic, non-standard spelling that the model has not seen. Normalising واہ variants would help.
6	FN	پاکستان پائندہ باد 💚PK شب بخیر ♥	Sincere style / label issue	A patriotic greeting with hearts and no context. It looks sincere, so the gold label is doubtful.
7	FP	"عوام عمران خان کی تمام سازشیں جان چکی ہے" … mention #GeoNews URL@	Platform cues / reported speech	A news headline with quotes, a mention, a hashtag and a URL. These cues are common in sarcastic posts, but here the quotes are reported speech.
8	FP	نون لیگ کے دلال … یہ … صحافی ہے؟	Rhetorical question	Direct abuse that ends in "؟" is read as rhetorical sarcasm. It is an insult and not irony.
9	FP	آج کا بہترین ایوارڈ یافتہ لطیف … بلاول بھٹو 🤣😂😆	Irony cue (praise + laughter)	Arguably sarcastic: positive wording with laughing emojis. The gold label is doubtful.
10	FP	ایک سوال … کیس نہی بنتا کیا ؟😉 … تسیں اپنے سمجھدار اوہ 😜	Code-mixing (Punjabi) + wink emojis	Punjabi wording with wink emojis and a rhetorical question reads as sarcastic, so the gold label is doubtful.

7. Reproducibility and acknowledgements

The notebook (urdu_sarcasm_pipeline_v4.ipynb) runs top to bottom on Google Colab with a T4 GPU, seed 42. Setup is described in README.md and requirements.txt. . Saved artifacts: fe_lr.joblib, tfidf_lr.joblib, bilstm_charcnn.pt, nn_vocab.json, results.csv, error_analysis.csv Libraries: scikit-learn, PyTorch, Hugging Face transformers and tokenizers, gensim, emoji, arabic_reshaper, python-bidi, pandas, matplotlib, seaborn. Pretrained model: xlm-roberta-large (Conneau et al., 2020). Urdu stopword list: Kaggle itsnobita/urdu-stopwords.

AI-assistance statement:

Data Visualization: Most of the data visualization has been completed, with AI used to a limited extent to improve the clarity and visibility of the visualizations.

Data Processing and Model Development: Most of the data processing was performed using Kaggle Notebooks. AI assistance was used to develop and experiment with neural network and Transformer architectures, including testing different hidden-layer configurations and evaluating different pretrained models.